

Correlation-Based Stint Filtering Biases Tyre Degradation Models: Evidence from Three Seasons of Formula 1 Telemetry

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Abstract—Tyre degradation models for Formula 1 conventionally estimate a per-stint degradation rate as the slope of fuel-corrected lap time against tyre age, and retain only stints whose linear fit exceeds a correlation threshold. We show that this near-universal quality criterion is not a quality criterion at all. Measured across 3 892 stints from the 2022–2024 seasons, the absolute correlation $|r|$ tracks slope *magnitude* ($\rho = 0.53$) far more strongly than it tracks slope *precision* ($\rho = -0.20$), because a genuinely flat stint has low correlation by construction however well it is measured. The resulting acceptance rate rises monotonically from 29.5% for near-zero slopes to 98.7% for large ones, systematically deleting low-degradation circuits from the training set. Replacing the criterion with one based on the standard error of the fitted slope raises the between-event share of target variance from 32.3% to 49.7%, the cross-season per-circuit correlation from 0.18 to 0.64, and corrects every remaining directional error in a downstream pit-strategy simulator. We further report that a telemetry-derived energy proxy widely treated as an aerodynamic load term correlates *negatively* with degradation ($\rho = -0.60$), falsifying the frictional-work hypothesis in that form, and argue from a variance decomposition that it is better understood as a measure of straight-line fraction. All results are accompanied by a measured performance ceiling, which we argue should be established before any improvement threshold is fixed.

Index Terms—motorsport analytics, tyre degradation, selection bias, time-series cross-validation, gradient boosting, Formula 1

I. INTRODUCTION

Tyre degradation is the dominant controllable variable in Formula 1 race strategy. A pit stop costs roughly 20–25 s of track position, so the decision of when to stop is governed almost entirely by the rate at which lap time deteriorates with tyre age. Public telemetry, exposed through the FastF1 library, has made this problem accessible outside the teams themselves, and a standard modelling recipe has emerged: aggregate laps into stints, fit a linear model of fuel-corrected lap time against tyre age, and treat the resulting slope as the degradation rate.

That recipe contains a filtering step that has, to our knowledge, not been examined. Because a fitted slope may be noisy, stints are retained only when the regression achieves some minimum absolute correlation, typically $|r| \geq 0.3$. The stated justification is fit quality. We show the criterion does not measure fit quality, and that its failure mode is not random but systematically biased against precisely the circuits where a strategist most needs an accurate answer.

The contributions of this paper are:

- 1) A quantitative demonstration that correlation-based stint filtering selects on slope magnitude rather than estimation precision, with an acceptance rate varying by a factor of 3.3 across the range of true slopes (Section IV).
- 2) A replacement criterion based on the standard error of the fitted slope, and measurement of its effect on variance structure, cross-season transferability and downstream strategy correctness (Section V).
- 3) Evidence that a commonly used energy proxy, $\overline{v^2}$, correlates negatively with degradation and is better interpreted as a track geometry descriptor than as a contact-patch work term (Section VI).
- 4) An argument, with worked example, that improvement thresholds should be set only after the achievable ceiling has been measured (Section VII).

II. RELATED WORK

Rubber friction is viscoelastic, decomposing into adhesion and hysteresis components; Persson’s multiscale theory obtains friction by integrating viscoelastic losses over the surface roughness spectrum [1]. Wear literature descending from Archard establishes that wear rate is proportional to frictional work dissipated at the contact patch [2], which motivates including an energy term in any composite wear index.

Direct application of that physics requires a tyre force model. The Pacejka Magic Formula [3], [4] is the standard, but its coefficients are load-dependent and fitted to rig data that is proprietary for Formula 1 constructions. We therefore exclude it, following the same reasoning as prior public work, and use curvature-derived proxies computed from position telemetry.

For the learning component, gradient-boosted trees remain state of the art on medium-sized tabular data, with inductive biases robust to uninformative features and irregular target functions [5], [6]. On validation, standard k -fold cross-validation leaks future information when observations are serially dependent [7], [8], requiring chronological splits. Prior Formula 1 work using FastF1 includes tyre energy forecasting with gradient boosting and recurrent models [12] and a state-space treatment separating fuel mass from latent tyre pace [13]; both note that the practical industry standard remains simple linear degradation models.

Derivative estimation from noisy telemetry uses Savitzky–Golay filtering [9], whose original coefficient tables contain typographical errors corrected subsequently [10]; we rely on a library implementation rather than transcribed coefficients.

III. DATA AND METHODOLOGY

A. Collection

We collect all race sessions from the 2022–2024 Formula 1 seasons, a single ground-effect regulation era, via FastF1. This yields 68 sessions and 49 993 laps. The API enforces a limit of 500 requests per hour. We measure the true cost per session before bulk collection: a race loaded with laps and weather but without telemetry costs 11 requests, a repeat load of the same session costs 0, and enabling telemetry adds 2.

The zero-cost repeat is architecturally important. The cached session class places the cache lookup ahead of the rate-limited transport in the method resolution order, so a cache hit never reaches the limiter. Preserving the cache therefore makes all subsequent processing free, and we write each session to its own file through an atomic rename so that interruption costs one session rather than the corpus.

B. Target construction

Lap times are fuel-corrected before any slope is fitted. Fuel remaining on lap N of a race of L laps, starting from the regulation maximum $F_0 = 110$ kg, is

$$F(N) = F_0 - (N - 1) \frac{F_0}{L}, \quad (1)$$

since the car completes lap N carrying the fuel it held at the start of that lap. The corrected time is $t^* = t - \kappa F(N)$ with $\kappa = 0.032$ s/kg. Omitting this correction produces a target dominated by fuel burn rather than tyre wear.

Representative laps are selected by a mask excluding in-laps and out-laps, non-green track status, the first lap of each stint, laps slower than 107% of the stint median, and laps without a recorded time. This retains 84.8% of laps.

For each stint s , defined by the tuple (season, round, driver number, stint number), we fit

$$t_s^*(a) = \beta_{0,s} + \beta_{1,s} a + \varepsilon, \quad (2)$$

where a is tyre age in laps. The degradation rate is $\beta_{1,s}$, in seconds of lap time lost per lap of tyre age.

C. Features and validation

Features must be knowable *before* a stint is run, since the intended application chooses stint length. Stint length and clean-lap count are therefore excluded as reverse-causal. The retained set comprises tyre age at stint start, compound as both an absolute Pirelli C-number and a within-weekend relative ordinal, stint mean air and track temperature, three circuit trait ratings, scheduled race distance, and constructor identity as a one-hot block.

Validation is chronological and grouped on whole events. Splitting rows rather than events allows stints from a single race to fall on both sides of a fold boundary, leaking shared

TABLE I
ACCEPTANCE RATE OF THE $|r| \geq 0.3$ CRITERION, BY TRUE SLOPE MAGNITUDE. A QUALITY CRITERION WOULD BE FLAT.

$ \hat{\beta}_1 $ (s/lap)	Stints	Accepted
0.00–0.02	505	29.5%
0.02–0.05	653	86.2%
0.05–0.10	935	98.7%
0.10–0.20	541	99.8%
> 0.20	179	99.4%

track, weather and safety-car conditions. We split on the ordered set of unique (season, round) keys and map back to rows, asserting that no event straddles any boundary. Models train on 2022–2023 and are evaluated on a held-out 2024 season.

IV. CORRELATION-BASED FILTERING IS MAGNITUDE SELECTION

The conventional criterion retains stint s when $|r_s| \geq 0.3$, where r_s is the Pearson correlation of the fit in (2). If this measures fit quality, $|r_s|$ should correlate with the precision of $\hat{\beta}_{1,s}$, that is inversely with its standard error $\text{se}(\hat{\beta}_{1,s})$, and should be independent of the magnitude $|\hat{\beta}_{1,s}|$.

We fit all 2815 dry stints without any filtering and measure both. The result is the opposite of the stated intent:

$$\rho(|r|, |\hat{\beta}_1|) = +0.53 \quad (3)$$

$$\rho(|r|, \text{se}(\hat{\beta}_1)) = -0.20 \quad (4)$$

The criterion tracks how large the degradation is roughly three times more strongly than how well it was measured. The mechanism is elementary: for a fit with residual scatter σ over n laps spanning tyre ages with standard deviation s_a ,

$$r \approx \frac{\beta_1 s_a}{\sqrt{\beta_1^2 s_a^2 + \sigma^2}}, \quad (5)$$

so $r \rightarrow 0$ as $\beta_1 \rightarrow 0$ regardless of how small σ is. A stint with genuinely no degradation cannot achieve high correlation, however precisely it is measured.

Table I shows the consequence. Acceptance rises monotonically from 29.5% to 98.7% across the range of true slopes.

The bias is not uniform across circuits. Monaco, the canonical low-degradation venue, has a median absolute slope of 0.0175 s/lap against 0.0614 elsewhere, but a median slope standard error of 0.0062 against 0.0152 elsewhere. Its stints are measured *more* precisely than average and were being discarded for being flat.

A second, compounding effect arises from the common practice of excluding negative slopes as physically implausible. Monaco’s true median slope is -0.008 s/lap. A circuit with no measurable degradation produces slopes scattered either side of zero, and truncating at zero retains only the positive half of that noise, biasing the estimate upward exactly where negligible degradation is the operationally important fact.

TABLE II
EFFECT OF REPLACING CORRELATION-BASED WITH PRECISION-BASED
FILTERING.

Quantity	$ r \geq 0.3$	$se \leq 0.02$
Between-event variance share	32.3%	49.7%
Oracle MAE ceiling	27.4%	38.0%
Cross-season circuit r (23 vs 24)	0.18	0.64
Holdout R^2 , gradient boosting	+0.047	+0.097
Holdout R^2 , linear baseline	-0.053	+0.044
Retained stints	2 101	1 726

V. A PRECISION-BASED CRITERION

We replace the correlation threshold with a bound on the standard error of the fitted slope, retaining stint s when

$$se(\hat{\beta}_{1,s}) \leq \tau, \quad \tau = 0.02 \text{ s/lap.} \quad (6)$$

The value of τ is chosen relative to the quantity of interest: a typical degradation rate is 0.065 s/lap, so τ admits stints whose rate is pinned well enough to distinguish a high-degradation circuit from a low one. This accepts a well-measured flat stint and rejects a noisy one, which is the property actually required.

The lower plausibility bound is set to -0.05 s/lap rather than zero, corresponding to approximately one second of improvement across a twenty-lap stint. This separates track evolution, which genuinely occurs, from measurement noise around a true rate of zero.

Table II reports the effect. Every quantity moves in the same direction, and the change is substantial rather than marginal.

The rise in cross-season correlation from 0.18 to 0.64 is the most consequential result. Under the correlation criterion, a circuit’s degradation in one season barely predicted the next, which would ordinarily be interpreted as genuine year-to-year instability arising from compound and car changes. It was instead an artefact of the filter. The signal was present throughout and was being removed.

We validate the correction against domain ground truth using a pit-strategy simulator driven by the resulting rates. Under the correlation criterion the simulator returned a two-stop strategy for Monaco, which is unambiguously wrong. Under the precision criterion it returns a one-stop for Monaco, Monza, Spa, Singapore, Jeddah and Silverstone, and a two-stop for Bahrain, Suzuka, Barcelona and Hungaroring, all consistent with observed practice.

VI. THE ENERGY PROXY MEASURES TRACK GEOMETRY

Physics-informed designs include an energy term in the target, motivated by the proportionality of wear to frictional work [2]. The standard telemetry proxy is $\overline{v^2}$, the mean squared speed over a stint, described as an aerodynamic load term.

We compute this from position telemetry over ten races (209 stints). Position coordinates are reported in units of 10^{-1} m rather than metres; since curvature scales as the inverse of length, neglecting this yields lateral accelerations an order of magnitude too small. After correction, the median peak lateral acceleration is 47.2 m/s^2 (4.8 g) and median peak combined

TABLE III
HELD-OUT PERFORMANCE, 2022–2023 TRAINING, 2024 EVALUATION.

Model	MAE (s/lap)	R^2
Mean predictor	0.03539	—
Linear baseline	0.03337	+0.044
Gradient boosting (tuned)	0.03345	+0.097

acceleration 4.85 g, both consistent with Formula 1 vehicle dynamics.

The measured correlation with degradation is strong and negative: $\rho = -0.596$ ($p = 1.9 \times 10^{-21}$). This falsifies the frictional-work hypothesis in the form stated, which predicts a positive relationship. A telemetry-free approximation using only speed-trap readings yields $\rho \approx 0$, so the cheaper substitute produces a false negative and does not permit the hypothesis to be tested at all.

A variance decomposition explains the sign. 92.6% of the variance of $\overline{v^2}$ lies between circuits rather than between stints, so it is a track descriptor, not a per-stint quantity. Circuits with high mean speed spend proportionally less of the lap cornering, and cornering rather than speed loads the tyre. Barcelona, at 196 km/h mean, averages +0.117 s/lap of degradation; Jeddah, at 230 km/h, averages +0.019. The circuit-level correlation is $\rho = -0.717$ ($n = 9$).

Compared against hand-assigned circuit severity ratings, $\overline{v^2}$ achieves $\rho = -0.596$ where a subjective composite rating achieves +0.578: comparable strength, opposite sign. We therefore interpret the proxy as an automatically measured circuit severity descriptor rather than a contact-patch work term, and suggest that the name *aerodynamic load proxy* is misleading.

VII. MEASURE THE CEILING BEFORE SETTING THE THRESHOLD

Our pre-registered acceptance criterion required the tuned model to improve on a linear baseline by at least 15% in held-out mean absolute error. It does not: the improvement is -0.2% .

That figure is uninterpretable without a ceiling. We compute an oracle predictor that assigns each stint the mean degradation of its own event and compound, which requires knowing the answer in advance and therefore bounds any model using event-level features. It improves on the global mean by 38.0%. The 15% threshold thus demanded 40% of all achievable headroom, a requirement fixed before anyone established what was achievable.

The reason for the low ceiling is structural: only 49.7% of degradation variance lies between events. The remainder separates stints within a single race, reflecting traffic, fuel management and driver instruction, none of which is recoverable from circuit-level features.

Table III shows the tuned tree ensemble level with the linear baseline on absolute error while achieving more than double its R^2 . Once the target’s construction bias is removed, the relationship between circuit severity and degradation is close

TABLE IV
CAPACITY SWEEP. THE OPTIMUM IS INTERIOR, BRACKETED BY UNDER-
AND OVER-FITTING.

Trees	Depth	Train MAE	Held-out MAE	Gap
20	2	0.03527	0.03496	-0.9%
50	3	0.03186	0.03388	+6.4%
71	5	0.02806	0.03334	+18.8%
150	5	0.02450	0.03379	+37.9%
400	4	0.02223	0.03518	+58.2%
1000	8	0.01451	0.03544	+144.3%

to linear, and the additional flexibility of boosting contributes only in the tails.

A. Fit adequacy

We verify that this result reflects the problem rather than a mis-specified model, via a capacity sweep (Table IV). Held-out error traces a clear minimum at 71 trees of depth 5. Below it, training and held-out error are both high and nearly equal, the signature of insufficient capacity; above it, training error falls to 0.0145 while held-out error rises, the signature of memorisation. The selected model sits at the interior optimum. The training-to-held-out gap is +14.7%, and the cross-validation estimate lies within 5.0% of held-out performance, indicating that the grouped chronological scheme is an honest predictor of generalisation.

Split-half reliability of the target, obtained by fitting (2) separately on odd and even clean laps of each stint, is 0.92 (Pearson). The target is therefore measured reliably; the difficulty lies in transferring across events, not in measuring within them.

VIII. DISCUSSION

A. Threats to validity

The precision threshold τ is itself a filter on the target and cannot be entirely free of selection effects. We chose it by reference to the magnitude of a typical degradation rate rather than by optimising held-out performance, and we note that a stricter correlation threshold was tested, produced a better R^2 , and was rejected on the grounds that its justification would have been its effect on the metric.

Retention falls from 59.7% to 44.3% under the precision criterion. The trade is favourable on every measured axis, but it is a trade.

Residuals correlate -0.26 with position through the held-out season ($p = 2.4 \times 10^{-11}$), indicating that the model is well fitted but not uniformly calibrated across a season. We report this rather than correct it, as any correction would consume the held-out set.

The energy-proxy analysis rests on ten races and nine circuits. The stint-level correlation is robust ($n = 209$, $p < 10^{-20}$), but the circuit-level figure ($n = 9$) is indicative only.

B. Practical implication

Because per-circuit degradation transfers across seasons at $r = 0.64$ – 0.74 once the bias is removed, while the model captures only 14.4% of achievable headroom, a strategy system should prefer directly observed per-circuit rates where they exist and use a model only to fill gaps. We adopt this ordering, and record for each estimate whether it is observed or modelled.

IX. CONCLUSION

A correlation threshold applied to a regression slope selects for large slopes, not for well-estimated ones. In tyre degradation modelling this systematically removes low-degradation circuits, and we have shown it suppresses cross-season transferability by a factor of three and produces demonstrably wrong strategy recommendations. Replacing it with a bound on the slope standard error is a one-line change that recovers the signal.

We further find that the standard $\sqrt{v^2}$ energy proxy correlates negatively with degradation and is more accurately described as a measure of track straight-line fraction, and we argue that improvement thresholds are not interpretable until the achievable ceiling has been measured.

The broader methodological point is that each of these errors was embedded in a design that was physically motivated, properly cited, and internally coherent. None was detectable by inspection; each required measuring the quantity that the design assumed.

REPRODUCIBILITY

All results are reproducible from the accompanying implementation. Collection is resumable and cached; the complete pipeline from cold API to trained model runs in approximately ten minutes of network time and minutes of computation on a single CPU core. Tuned hyperparameters are persisted and version-checked against the feature set.

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